

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1043

Ibm Cottle Tap 100 kV
August 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1043. The Interconnection Customer proposes a 27.1 MW solar photovoltaic facility interconnecting at the Ibm Cottle Tap 100 kV substation in the Indiana area. The request was received on August 25, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 276 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$195,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The State St-Waynedale 230.0 kV is overloaded to 109.0% of its 991 MVA emergency rating for the loss of the Richlad-Deklb 500.0 kV. The project contributes 17.3% DFAX (4.7 MW) to this overload.

Swan Vly-Twin County Ranch 230.0 kV Line: 103.4% loading (1,145 MVA) following the outage of the Richland-Troy Energy Llc 500.0 kV; DFAX = 15.5%.

The Wabash 500/230 kV XFMR is overloaded to 103.1% of its 1,666 MVA emergency rating for the loss of the Richland-Dekalb 500.0 kV L. The project contributes 19.7% DFAX (5.3 MW) to this overload.

George Neal S-Myrtle Pt 230.0 kV: 102.1% loading (1,211 MVA) following the outage of the Trego-Galion Generating Sta 500.0 kV line; DFAX = 6.4%.

For the loss of Richlnd-Dekalb 500.0 kV line, the 24Th & Gardendale-Landsman Creek 230.0 kV loads to 101.3% of its 832 MVA rating. Project DFAX: 8.6%.

The Sidney A Murray Jr Hydroelectric-24Th & Gardendale Substation 230.0 kV is overloaded to 100.8% of its 860 MVA emergency rating for the loss of the Dekalb-Richland 500.0 kV. The project contributes 8.6% DFAX (2.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Waynedale-State St 230.0 kV	Reconductor line	\$5,495,000	0% (\$11,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Swan Vly-Twin County Ranch 230.0 kV Line	Reconductor line	\$11,160,000	0% (\$35,000)
Wabash 500/230 kV XFMR	Replace transformer with higher-rated unit	\$35,753,000	0% (\$69,000)
Myrtle Pt-George Neal S 230.0 kV	Reconductor line	\$3,786,000	0% (\$7,000)
24Th & Gardendale-Landsman Creek 230.0 kV	Reconductor line	\$14,677,000	0% (\$47,000)
24Th & Gardendale-Sidney A Murray Jr Hydroelectric 230.0 kV	Reconductor line	\$8,270,000	0% (\$26,000)
	Total allocated to project		\$195,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.